Bracketing a root

CodEx — Exercices de programmation

Let $f(x) = -x^3 + x^2 + x + 1$. Write `dichotomie(debut, fin, precision)` to return a root bracket of width at most `precision`. Assume `debut < fin`, `precision > 0`, f continuous and strictly monotone on the interval, and $f(\texttt{debut})f(\texttt{fin}) < 0$. If a midpoint is an exact root, return that midpoint twice.

The requested precision must remain attainable with the floating-point numbers used.